A Machine Learning Excursion to Cellular Automata

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Abstract: In this project, we toy around with machine learning algorithms to predict the next generation of a cellular automaton. Specifically, we focus on Rule 90 and investigate how models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks can learn to make these predictions. We find that accurately predicting the next generation with RNNs and LSTMs is indeed a challenging task due to the inherent complexity and chaotic nature of Rule 90. We then have a look at Game of Life, through the lens of CNNs.

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1 Introduction

Cellular Automata, conceptualized by Stanislaw Ulam and John von Neumann in the 1940s, are fascinating toy systems where the emergence of complexity from simplicity can be explored. These systems, defined by discrete grids and simple local rules, serve as foundational models for understanding dynamic processes in mathematics, computation, biological phenomena and physics amongst other disciplines.

1.1 Wolfram and his automata

Stephen Wolfram revolutionized the study of CA with "A New Kind of Science". In essence, Wolfram advocates the notion that when simple code instructions are continually applied, it is possible to generate an astonishing range of complexities from order, through periodic functions all the way to complete chaos. Cellular automata, as a computing machine, represent this principle, which integrates the basic elements of computer modeling of scale in natural and artificial systems.

Wolfram's classification comprises one-dimensional cellular automata composed of a one-dimensional lattice of cells with 0 and 1 states, as examples. Each cell's state changes according to deterministic transition rules concerning the states of cells surrounding it. For three cells including itself and its two immediate neighbors, the set of binary CAs would have $2^{2^3} = 256$ possible configurations. These rules were also deemed as systems of rules

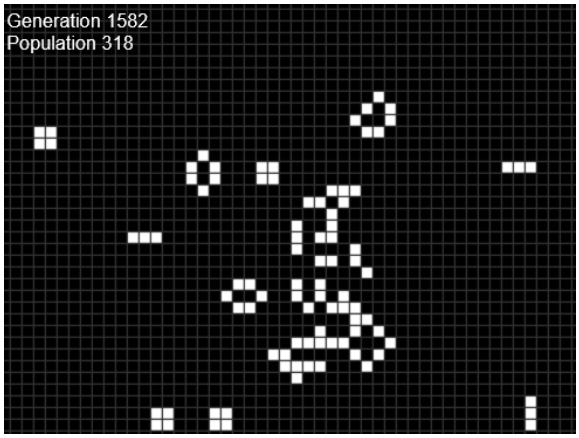

Figure 1: A Snapshot of Game of Life

and Wolfram distributes binary numbers from 0 to 255 for each of these rules based on the output of the rule.

1.2 Rule 90

Among Wolfram's classifications, Rule 90 stands out as a fascinating example of simplicity yielding complexity. Rule 90 is a one-dimensional, binary cellular automaton where the state of each cell at the next time step is determined by the exclusive OR (XOR) of its two immediate neighbors' states. Martin, Odlyzko and Wolfram call it "the simplest non-trivial cellular automaton". Mathematically, the rule is represented as:

$$s_i^{t+1} = s_{i-1}^t \bigoplus s_{i+1}^t$$

1.3 Game of Life

Conway's Life is a two-dimensional cellular automaton with a simple local update rule that can produce complex global behavior. In a Life configuration, cells in an n $\times$ n grid can be either alive or dead (represented by 1 or 0 respectively). To determine the state of a given cell on the next step, Life considers the 3 $\times$ 3 grid of neighbors (Moore's neighborhood) around the cell. Every step, cells with exactly two alive neighbors will maintain their state, cells with exactly three alive neighbors will become alive, and cells with any other number of neighbors will die. We consider a variant of Life in which cells outside of the n $\times$ n grid are always considered to be dead. Despite the simplicity of the update rule, Life can produce complex output over time.

1.4 Recurrent Neural Networks (RNNs)

RNNs are a type of neural network designed for sequential data, where the output at each step depends on the previous inputs. They have internal loops that allow information to persist, making them ideal for tasks like time series prediction, language modeling, and sequence classification. However, RNNs often face issues with vanishing or exploding gradients, which can limit their ability to learn long-term dependencies.

1.5 Long Short-Term Memory Networks (LSTMs)

LSTMs are an advanced type of RNN that address the vanishing gradient problem by introducing a memory cell and three gating mechanisms: input, output, and forget gates. These gates regulate the flow of information, allowing LSTMs to retain important features from long sequences. This makes them particularly effective for tasks like speech recognition, text generation, and video analysis.

1.6 Convolutional Neural Networks (CNNs)

CNNs are specialized neural networks primarily used for processing grid-like data, such as images. They consist of convolutional layers that apply filters to extract spatial features, followed by pooling layers to reduce dimensionality. CNNs excel at tasks like image classification, object detection, and visual recognition due to their ability to learn hierarchical patterns in data.

2 Methodology

2.1 Rule 90

We use RNN and LSTM network to predict the next generation given the previous one. The parameters we used are in the table. We used adam optimizer, and binary cross entropy as our loss function in both RNN and LSTM data sets. We also played with CNNs though the results are not included in the report.

Parameters	Values
sequence_length	20
num_samples	10000
embedding_dim	8
nn_units	32
batch_size	32

Table 1: Parameters used for experimentation.

2.2 Game of Life

We use a simple CNN model to predict the next generation of a configuration of Game of Life. The convolution layer has only one filter of the size 3×3, and apply it all over our Life grid, apply the sigmoid activation, and keep the same dimensions in the output due to padding. Adam optimizer and cross entropy loss function were used. We generate 10,000 samples, and vary grid size. The grid sizes used were 20, 50 and 100. An auto cutoff for training was also coded in the model. We used 'early stopping' on the loss with a patience 3.

3 Experimentation and Results

3.1 Rule 90

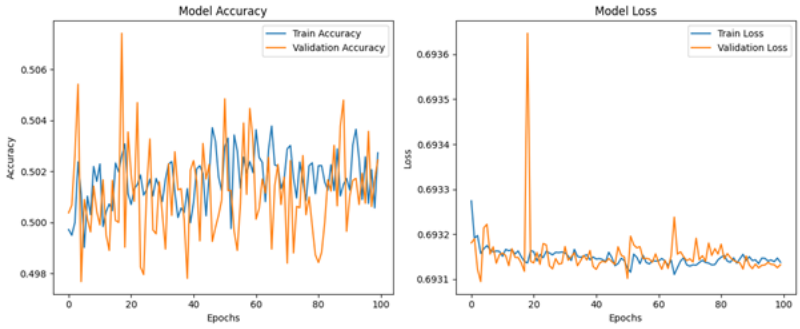

Figure 2: RNN, trained for 100 epochs.

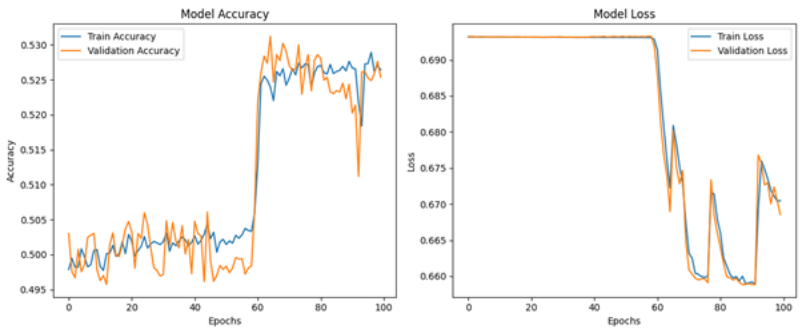

Figure 3: LSTM, trained for 100 epochs.

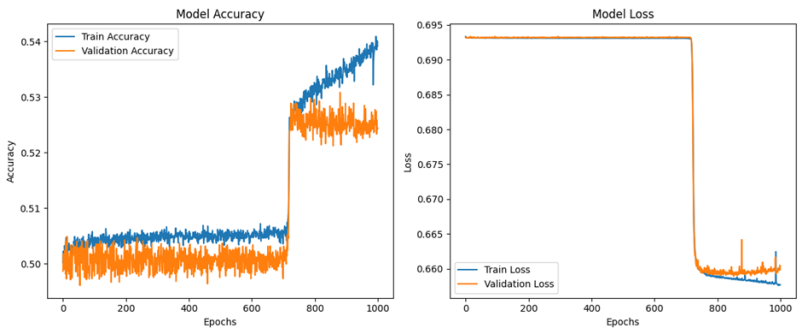

Figure 4: RNN, trained for 1000 epochs.

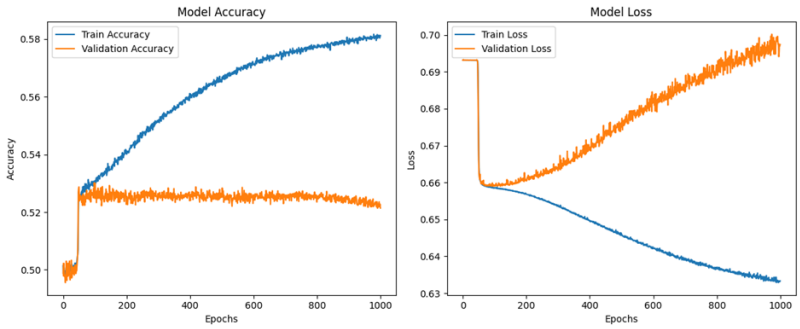

Figure 5: LSTM, trained for 1000 epochs.

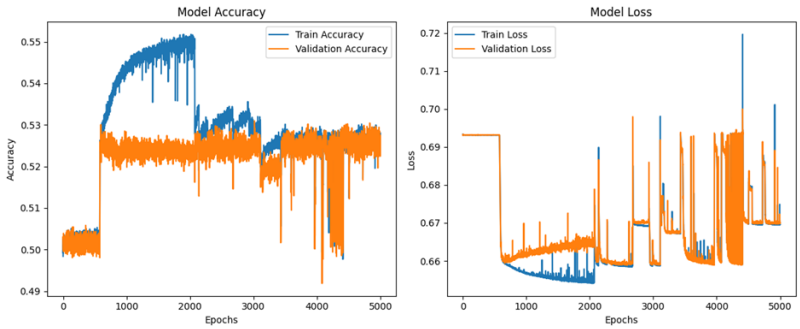

Figure 6: RNN, trained for 5000 epochs.

3.2 Game of Life

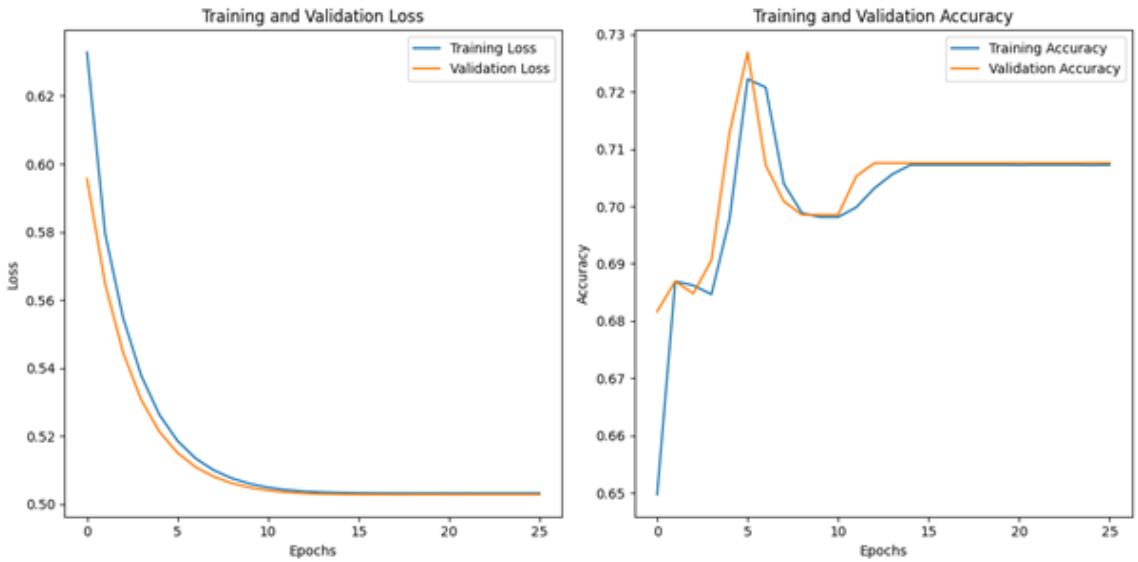

Figure 7: Statistics for Grid Size 20.

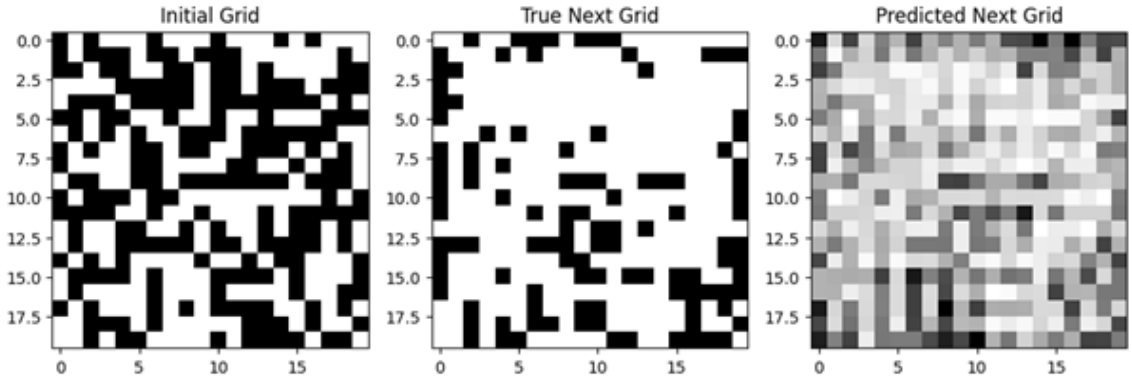

Figure 8: Predicted outcomes for Grid Size 20.

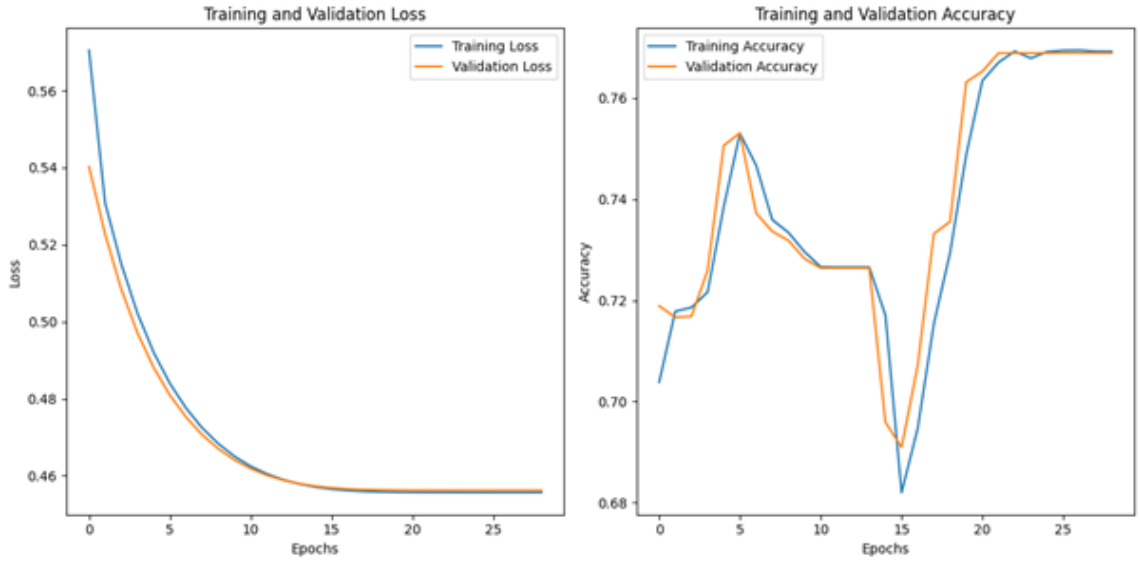

Figure 9: Statistics for Grid Size 50.

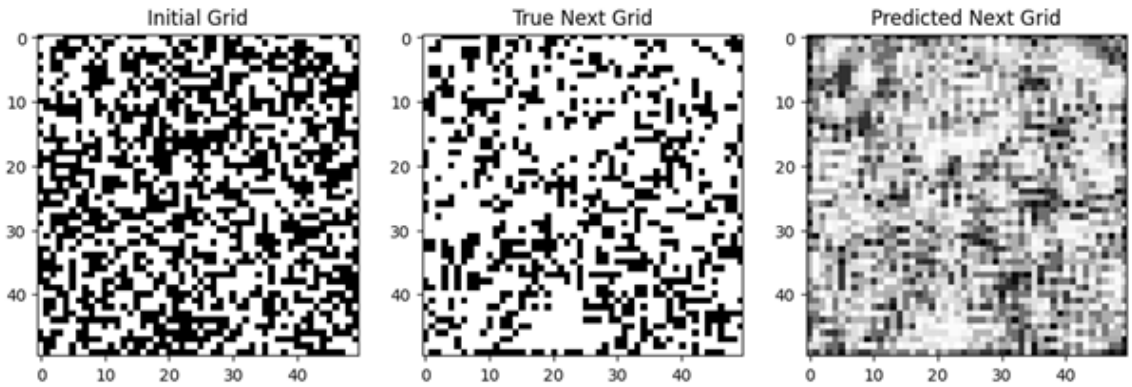

Figure 10: Predicted outcomes for Grid Size 50.

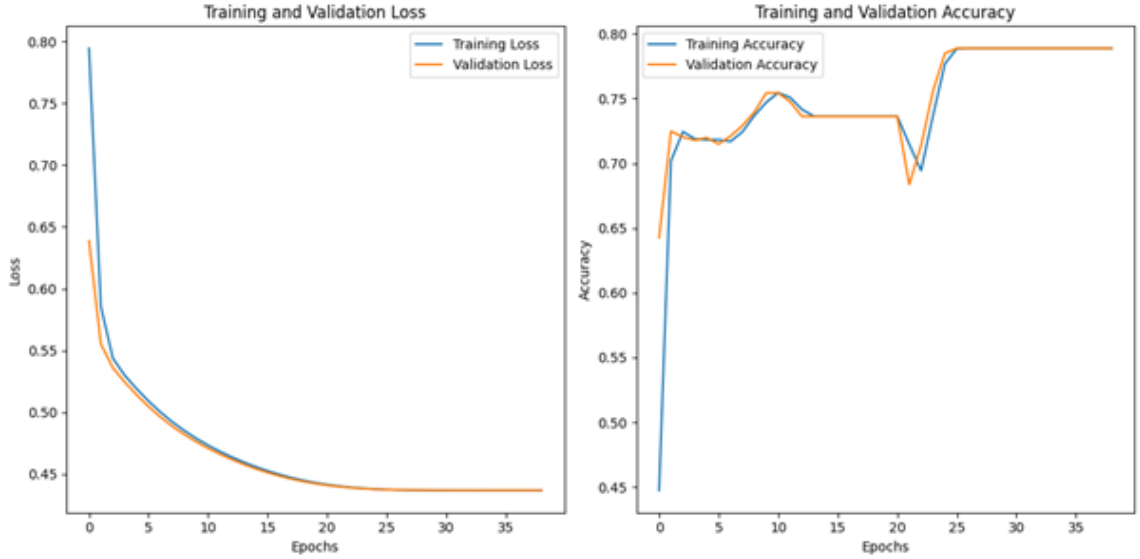

Figure 11: Statistics for Grid Size 100.

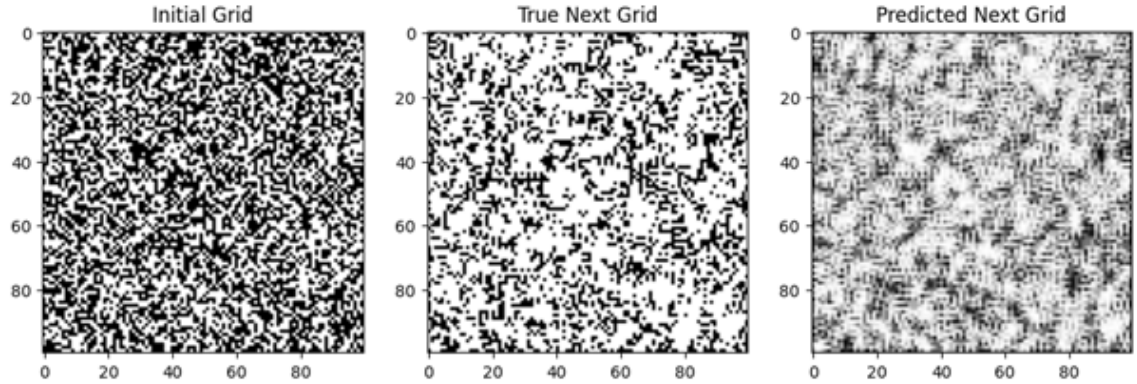

Figure 12: Predicted outcomes for Grid Size 100.

4 Conclusion

We find that RNNs, and LSTMs do not perform too well when dealing with the problem of generating the next generation of Rule 90, given some configuration. It is curious to find out that we weren't able to get more than about 53% accuracy when using RNN or LSTM. They both seem to have similar behavior but scaled on an epochal level. CNNs give pretty good accuracy, which is expected as they probe the local relationships of the data, and might be able to find the XOR relation. The Game of Life results are more interesting. We find that with increasing grid size, the accuracy of prediction increases, with the predicted grid of size of 100 resembling the most with the true next generation. Learning CA, still seems to be difficult task for CNN.

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